

SENTHIL KUMAR NAGAPPAN

AI Safety & Experience Design · Human Systems Integration · UX/UI & Human Factors · Accessibility Leadership

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U.S. Citizen · Active Public Trust Clearance · TS/SCI Eligible

PROFILE

I design how people and AI work together — safely, clearly, and accessibly.

For 18+ years I have led the design of intuitive, trustworthy, human-centered experiences for healthcare, federal, and regulated environments — the places where getting the experience wrong has real consequences. I work at the intersection of four disciplines most organizations keep separate: AI experience design, human factors engineering, UX/UI design, and front-end development — turning complex systems, dense data, and unpredictable AI into products that are clear, usable, and accountable.

My recent work centers on AI safety and experience — designing transparency, trust, explainability, and human-in-the-loop oversight into AI-driven products in clinical settings. Alongside that, I lead enterprise accessibility programs (WCAG 2.2 AA, Section 508) and apply medical-device human factors standards (IEC 62366, ISO 14971, FDA HFE) where usability is a safety requirement, not a nicety.

WHAT I DO

AI Experience & Safety — AI experience design, conversation design, explainability and confidence indicators, human-in-the-loop oversight, hallucination/fallback handling, responsible-AI patterns, system-prompt design

UX/UI & Research — User-centered design, generative and evaluative research, usability testing, wireframes, prototypes, high-fidelity UI, design systems, journey mapping, service design, interaction design

Human Factors & HSI — Human factors engineering, human systems integration, task analysis, use-error risk analysis, usability validation, IEC 62366, ISO 14971, AAMI HE75, FDA HFE, MIL-STD-1472H

Accessibility — Enterprise accessibility programs, WCAG 2.2 AA, Section 508, audits, remediation, VPAT/ACR, governance, inclusive design, assistive tech (JAWS, NVDA, VoiceOver) — IAAP CPACC + WAS

Front-End — React, TypeScript, JavaScript, HTML5, CSS3, Redux, responsive design, design-to-dev handoff, automated testing (Jest, Cypress), AI-assisted design (Figma, Lovable)

SELECTED WORK

AI + Human Systems Integration Lead | GE HealthCare (Brain Pulse Inc)

2025 – Present

Establishing an AI safety framework for clinical decision-support and prenatal diagnostics

- **Established the AI safety framework** for clinical decision-support — model evaluation criteria, deployment thresholds, and risk controls now used as an enterprise standard for how AI is designed, validated, and released.
- **Designed trustworthy AI experiences** — transparency, confidence indicators, and human-in-the-loop oversight for safety-critical clinical AI — with conversation design and responsible-AI patterns aligned to FDA HFE, IEC 62366, and ISO 14971.
- **Led human factors and HSI across 10+ teams** — task analysis, use-related risk analysis, and usability validation embedded in regulated medical-device development.

Senior UX & Accessibility Lead — SSA11y | Social Security Administration (Leidos)

2022 – 2025

An AI-powered accessibility platform on the Disability Case Processing System, serving 50M+ citizens

- **Built SSA11y** — an AI platform (Llama2) with a React/TypeScript front-end — transforming accessibility from manual audits into continuous AI-driven validation, saving \$1.5M+ annually and cutting audit effort 30%.
- **Designed complex case-processing workflows** for disability adjudication — reducing cognitive load and error for caseworkers handling high-stakes decisions across 15+ modules.
- **Uncovered that 65% of accessibility issues came from just 8 shared components** — reshaping the design system and eliminating 200+ violations to WCAG 2.2 AA and Section 508 for 50M+ users.

UX & Human Factors Lead — Connected Health | Best Buy Health

2021 – 2022

Designing connected medical and emergency-response devices for 1M+ older adults

- **Led UX and human factors for connected medical devices** (Lively, Jitterbug) — patient-monitoring and safety-critical interactions for 1M+ users aged 65–90+, applying IEC 62366, FDA HFE, and MIL-STD-1472H.
- **Designed for low vision, dexterity, and cognitive-load constraints** — native iOS and Android, reducing critical interaction failures 30–40% across 5 product lines.

Senior UX Designer & Program Lead — TEDS & Behavioral Health | HHS / SAMHSA

2014 – 2021

(Eagle Technologies)

A \$130M behavioral-health portfolio, including the TEDS national data system, reaching 2M+ monthly

- **Led UX and accessibility across a \$130M federal portfolio** — 6+ behavioral-health platforms serving 2M+ monthly visitors over a 7-year engagement.
- **Designed the application behind the TEDS national database** — complex data-entry, validation, querying, and reporting interfaces for state administrators and federal analysts, plus data-visualization dashboards turning national datasets into actionable insight.

Earlier: Lead UX/UI Designer at AAMC · Sr. UX & Accessibility Consultant at CareFirst BCBS · UX & Front-End Designer at JPMorgan Chase · UX Consultant at Morgan Stanley & Threewide Corp.

EDUCATION, CERTIFICATIONS & AFFILIATIONS

B.E., Electronics & Communication Engineering — Madurai Kamaraj University, 1999

CUA (Certified Usability Analyst — HFI) · **CPACC + WAS** (IAAP — Accessibility) · **CSM** (Scrum Alliance) · **DAU HSI** (In Progress)

Industries: Healthcare & MedTech · Federal Government · Financial Services · Clinical AI · Connected Devices · Accessibility & Regulated Environments

LEADERSHIP & COMMUNITY

Vice President, Texas Astronomical Society (500+ members) · **Cub Master / Asst. Scoutmaster**, Scouting America · **Lead Coach**, FIRST Lego League — mentoring 150+ youth in STEM and the engineering design process.